MATHS 7105 Data Literacy

Assignment 1

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Task 1. Ghostbusters

The effect of negative reinforcement on ESP ability.

Description

The video shows a survey being conducted in which the interviewer has 2 participants(Boy and Girl), Both of them are asked to guess the shapes or whatever is on the card without looking at it. The boy is being repetitively told that he is wrong even when he manages to guess a correct answer while the girl in the video gives all the wrong answers and yet is told that she guessed all the answers correctly. The interviewer uses negative reinforcement(i.e. electric shock) on the boy every time he is told that his answer is incorrect, to test the ‘Effects of negative reinforcement on ESP ability’ which results in the boy getting frustrated and leaving the survey midway.

The negative reinforcement used in the video also shows the discomfort that the participant is facing and questions the ethics of this survey as well as the credibility of the data used for sampling.

Issue 1 – Credibility of the Data.

Sometimes Personal interviews might have an issue, and that is the interviewers themselves. If [a1895211]the interviewer fails to jot down the responses or deliberately tampers the original responses given by the participant it may result in biased or incorrect data. For instance, in the video when the boy does give a correct answer the interviewer lies and tells the participant that he is wrong and gives him a shock again by which he gets frustrated and leaves, also the interviewer's facial expression and responses towards the girl were positive and he kept praising her even after receiving incorrect answers from her, which raises questions on the accuracy of the data given the clear biases visible in the video.

Issue 2 – Selective Discrimination

According to APA’s ethics code, ‘It is ethically essential for the interviewer/psychologist not to engage in any kind of discriminating tasks in their work-related tasks regardless of any reasons’ (American Psychological Association. (n.d.). 3.01 Informed Consent to Research. APA Ethics Code. https://www.apa.org/ethics/code). While it was seen in the video that the boy was being electrocuted on every response(as a penalty for answering incorrectly) regardless of him being correct or not, the girl was never once given a penalty for answering incorrectly and shows a level of discrimination and errors while recording the responses.

Issue 3 - Informed Participation acknowledgement.

Obtaining informed consent from participants and their acknowledgment is essential as it informs participants about the purpose, duration, procedures, the option to decline the research, possible benefits and the discomforts as well as incentives if any(American Psychological Association. (n.d.). 8.02 Informed Consent to Research. APA Ethics Code. https://www.apa.org/ethics/code).

While it was clear by the male participant's statements given in the video that there was no acknowledgment or informed consent taken from the participants about his/her option to decline the research once it started neither was he told about receiving penalties for giving false answers beforehand which in this case were electric shocks.

Suggested new research method

1. Statement of objectives. The participant's informed consent and acknowledgment for the procedures need to be an essential step where purpose of the research is clearly stated accompanying every other info needed to be known.
2. Target population. The target population has to be clearly defined for selecting proper individuals.
3. The frame. The frame should be close to the Target population specified.
4. Sample Design. Have the sample properly and accurately designed and follow it throughout the survey.
5. Method of measurement. The method of measurement has to be equal for all.
6. Measurement Instrument. Make sure the measurement instrument is not biased as it may produce incorrect data.
7. Selection and training of field workers. Adequate training is essential so that less error occur while recording the responses.

Biblography

Extrasensory Perception (book) - Wikipedia

Scheaffer, Menden-hall & Ott 2006, Ch. 2, "Elements of the sampling problem" pp. 2.4–2.6, pp.18–38.

Ethical principles of psychologists and code of conduct (apa.org)